Sunscreen

MATH 213

Group 1

Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.

1. What are the cases? Participants in experiment
2. What is the explanatory variable? Sunscreen use (per ounce?)
3. What is the response variable? Skin cancer rate (severity of cancer if has)

Reminder of description: *Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.*

4. Does this mean that sunscreen *causes* cancer? Discuss with your partner and summarize your discussion below.

While the data may suggest that, it is impossible to say for absolute certain, as correlation is not causation, and there could be other factors that simply weren't taken into account. The data may suggest it, but it's not for certain.

Group 2

Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.

1. What are the cases? The participants
2. What is the explanatory variable? Level of sunscreen used (0 = none, 1 = normal, 2 = extra)
3. What is the response variable? Has/had skin cancer

Reminder of description: *Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.*

4. Does this mean that sunscreen *causes* cancer? Discuss with your partner and summarize your discussion below.

We can’t say that sunscreen causes skin cancer, but it may be correlated to its rates.

Group 3

Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.

1. What are the cases? Participants
2. What is the explanatory variable? Do you use sunscreen regularly? (Yes/No)
3. What is the response variable? Have you gotten skin cancer? (Yes/No)

Reminder of description: *Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.*

4. Does this mean that sunscreen *causes* cancer? Discuss with your partner and summarize your discussion below.

No correlation does not always equal causation; just because an individual does or does not use sunscreen does not necessarily indicate that they will get skin cancer.

Group 4

Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.

1. What are the cases?
    - Participants
2. What is the explanatory variable?
    - Sunscreen usage
3. What is the response variable?
    - Having smmkkin cancer

Reminder of description: *Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.*

4. Does this mean that sunscreen *causes* cancer? Discuss with your partner and summarize your discussion below.\

Not necessarily - for example, there may be other factors such as family history of cancer. Also, a participants may use more sunscreen because they are out in the sun more often (which could be an explanatory variable too)

Group 5

Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.

1. **What are the cases?**

The cases are all the participants of the research study.

2. **What is the explanatory variable?**

How much sunscreen a person uses.

3. **What is the response variable?**

The chance that person develops skin cancer.

Reminder of description: *Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.*

4. Does this mean that sunscreen *causes* cancer? Discuss with your partner and summarize your discussion below.

No, not necessarily. There may be unknown bias that caused the data to imply this or there may be not enough cases to give a clear answer. There is also the possibility that sunscreen does not largely impact the chance to develop skin cancer and that the ones who used more sunscreen just got unlucky.

Group 6

Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.

1. What are the cases?
2. What is the explanatory variable?
3. What is the response variable?

Reminder of description: *Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.*

4. Does this mean that sunscreen *causes* cancer? Discuss with your partner and summarize your discussion below.

Group 7

Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.

1. What are the cases?

   The # of people using sunscreen

2. What is the explanatory variable?

   Sunscreen usage, 1 to 5, how often do you use sunscreen when you go out.

3. What is the response variable?

   Did person get skin cancer?

Reminder of description: *Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.*

4. Does this mean that sunscreen *causes* cancer? Discuss with your partner and summarize your discussion below.

Based off of one study, sunscreen does not inherently cause cancer because it is known that people have used it and have not gotten cancer.

Group 8

Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.

1. What are the cases?
2. What is the explanatory variable?
3. What is the response variable?

Reminder of description: *Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.*

4. Does this mean that sunscreen *causes* cancer? Discuss with your partner and summarize your discussion below.

Group 9

Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.

1. What are the cases? Participants of the study

2. What is the explanatory variable? Do you use sunscreen (yes/no or sliding scale)? [sunscreen usage unclear]

Better, like how often do you use sunscreen?

3. What is the response variable? ~~Likelihood of having~~ Have you had skin cancer (yes/no)?

How many times have you had skin cancer? [needs to be measurable]

Reminder of description: *Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.*

4. Does this mean that sunscreen *causes* cancer? Discuss with your partner and summarize your discussion below.

No

Other factors, like UV exposure and SPF, sun exposure, people more prone to skin cancer will likely use sunscreen more often

Group 10

Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.

1. What are the cases?
    - Participants
2. What is the explanatory variable?
    - Cases of skin cancer per # of participants (Skin cancer diagnosis past or present)
3. What is the response variable?
    - Sunscreen Usage (Bottles purchased per month)

Reminder of description: *Suppose that researchers conduct a study that records sunscreen use and skin cancer for the participants. Suppose they find that the more sunscreen someone used, the more likely the person was to have skin cancer.*

4. Does this mean that sunscreen *causes* cancer? Discuss with your partner and summarize your discussion below.

The higher likelihood of cancer means that the patients are going to take proactive actions to protect against cancer by using sunscreen.